

Beef in an Optimal Lean Diet study: effects on lipids, lipoproteins, and apolipoproteins

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Background: A Step I diet with lean beef compared with lean white meat both decrease LDL cholesterol. To our knowledge, no studies have evaluated a low-saturated fatty acid (SFA) (<7% calories) diet that contains lean beef. Objective: We studied the effect on LDL cholesterol of cholesterol-lowering diets with varying amounts of lean beef [ie, Dietary Approaches to Stop Hypertension (DASH): 28 g beef/d; Beef in an Optimal Lean Diet (BOLD): 113 g beef/d; and Beef in an Optimal Lean Diet plus additional protein (BOLD+): 153 g beef/d] compared with that of a healthy American diet (HAD). Design: Thirty-six hypercholesterolemic participants (with LDL-cholesterol concentrations >2.8 mmol/L) were randomly assigned to consume each of the 4 diets (HAD: 33% total fat, 12% SFA, 17% protein, and 20 g beef/d), DASH (27% total fat, 6% SFA, 18% protein, and 28 g beef/d), BOLD (28% total fat, 6% SFA, 19% protein, and 113 g beef/d), and BOLD+ (28% total fat, 6% SFA, 27% protein, and 153 g beef/d) for 5 wk. Results: There was a decrease in total cholesterol (TC) and LDL-cholesterol concentrations ($P < 0.05$) after consumption of the DASH (-0.49 ± 0.11 and -0.37 ± 0.09 mmol/L, respectively), BOLD (-0.48 ± 0.10 and -0.35 ± 0.9 mmol/L, respectively), and BOLD+ (-0.50 ± 0.10 and -0.345 ± 0.09 mmol/L, respectively) diets compared with after consumption of the HAD (-0.22 ± 0.10 and -0.14 ± 0.10 mmol/L, respectively). Apolipoprotein A-I, C-III, and C-III bound to apolipoprotein A1 particles decreased after BOLD and BOLD+ diets compared with after the HAD, and there was a greater decrease in apolipoprotein B after consumption of the BOLD+ diet than after consumption of the HAD ($P < 0.05$ for both). LDL cholesterol and TC decreased after consumption of the DASH, BOLD, and BOLD+ diets when the baseline C-reactive protein (CRP) concentration was <1 mg/L; LDL cholesterol and TC decreased when baseline CRP concentration was >1 mg/L with the BOLD and BOLD+ diets. Conclusions: Low-SFA, heart-healthy dietary patterns that contain lean beef elicit favorable effects on cardiovascular disease (CVD) lipid and lipoprotein risk factors that are comparable to those elicited by a DASH dietary pattern. These results, in conjunction with the beneficial effects on apolipoprotein CVD risk factors after consumption of the BOLD and BOLD+ diets, which were greater with the BOLD+ diet, provide support for including lean beef in a heart-healthy dietary pattern. This trial was registered at clinicaltrials.gov as NCT00937898.